

Exercise Report

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Data Analytics in Finance
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1 Background

All of this work should be done within a Jupyter notebook. Please format it consistently with the notebooks that we used during the course. In particular, place emphasis on making your comments, code, and plots readable. Turn in both the .ipynb file and a .pdf version.

2 Part I

Return to Question Two of EX1. Refit the model with all of the predictors, but first transform each predictor on the ratio scale using the Yeo-Johnson transformation. (The objective here is to give each predictor distribution more of a "normal" shape.) Then, refit the model, but use the Box-Cox procedure to find the optimal transformation of the response. Show the results of the resulting model fit, including the plot of residuals versus fitted values.

2.1 Solution

2.1.1 Understanding the Why?

The transformation parameters λ_y (Box-Cox) and λ_j for each predictor (Yeo-Johnson) are selected via maximum likelihood or normality-optimizing criteria. After transformation, standard regression diagnostics (e.g., residual plots) are used to validate model assumptions. The Box-Cox transformation stabilizes variance and corrects skewness in the response. The Yeo-Johnson transformation adjusts predictors to better satisfy the linear model assumptions even when they are zero or negative. Together, these transformations improve the quality and interpretability of the linear regression fit.

2.2 Mathematical Proof

Let $y \in \mathbb{R}^n$ be the response vector, and $X \in \mathbb{R}^{n \times p}$ the matrix of predictor variables. We seek to transform the data to improve linear regression model assumptions such as linearity, homoscedasticity, and normality of errors. The Box-Cox transformation is defined for $y_i > 0$ as:

$$y_i^{(\lambda)} = \begin{cases} \frac{y_i^\lambda - 1}{\lambda}, & \lambda \neq 0 \\ \log y_i, & \lambda = 0 \end{cases}$$

where $\lambda \in \mathbb{R}$ is the transformation parameter. The goal is to choose λ that maximizes the log-likelihood of the transformed response under a normal error model. The Yeo-Johnson transformation generalizes Box-Cox to allow zero or negative values, and is defined for each predictor $x \in \mathbb{R}$ as follows.

$$T(x; \lambda) = \begin{cases} \frac{[(x+1)^\lambda - 1]}{\lambda}, & x \geq 0, \lambda \neq 0 \\ \log(x + 1), & x \geq 0, \lambda = 0 \\ \frac{-[(-x+1)^{2-\lambda} - 1]}{2-\lambda}, & x < 0, \lambda \neq 2 \\ -\log(-x + 1), & x < 0, \lambda = 2 \end{cases}$$

Each feature column in X is transformed using a fitted λ_j that optimizes normality for that feature. After transformation, the model becomes the following.

$$y^{(\lambda_y)} = X^{(\lambda_x)}\beta + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 I)$$

where $y^{(\lambda_y)}$ is the Box-Cox transformed response and $X^{(\lambda_x)}$ is the matrix of Yeo-Johnson transformed predictors β is estimated using ordinary least squares (OLS) as follows.

$$\hat{\beta} = (X^{(\lambda_x)\top} X^{(\lambda_x)})^{-1} X^{(\lambda_x)\top} y^{(\lambda_y)}$$

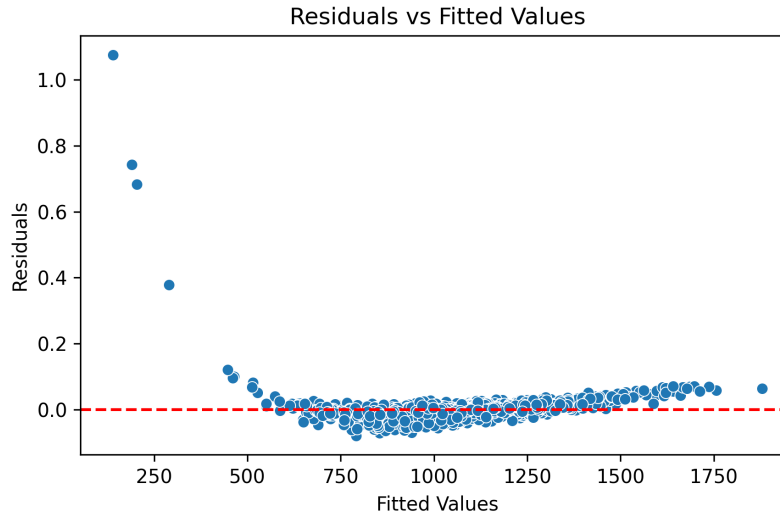


Figure 1: Residuals vs Fitted Values After Yeo-Johnson and Box-Cox Transformations

2.3 Summary

To address skewness and improve model assumptions, we applied transformation techniques targeting both predictors and the response. All predictors on a ratio scale were transformed using the Yeo-Johnson method, which generalizes the Box-Cox approach to allow for zero or negative values. This transformation tends to pull skewed distributions toward normality, potentially stabilizing variance and enhancing linear model performance. We used Box-Cox to optimally transform the response variable (`trade_price`). The estimated optimal lambda was 1.5951. This value, being close to 2, suggests a mild convex transformation (between

square root and quadratic). We refit the model using the Yeo-Johnson-transformed predictors and the Box-Cox-transformed response. The regression output revealed the following.

1. $R^2 = 1.000$, indicating potential overfitting or redundancy.
2. Many coefficients were highly significant (e.g., `trade_price`, `current_coupon`), while others showed reduced influence (e.g., `weight`, `reporting_delay`).
3. The smallest eigenvalue is extremely small (2.71^{-28}), signaling multicollinearity or near-singular design matrix structure.

We note that the residuals vs. fitted plot for the transformed model showed clear improvement in variance stabilization. However, slight curvature still remains, possibly from interaction terms or higher-order effects not captured in the linear framework. This dual-transformation approach helps restore the modeling assumptions of linear regression. While not perfect, it reduces skewness and stabilizes residual behavior. However, multicollinearity remains a concern, and further dimensionality reduction (e.g., PCA) or regularization (e.g., Ridge) could be considered next.

3 Part II

Let $X_1, X_2, \dots, X_n$ be i.i.d. from the Beta(a, b) distribution. Find numerically the maximum likelihood estimators for a and b . Mimic the function `gammamle()` that we created in lecture. The function should also return the covariance matrix for the MLE. Print the function that you wrote and attach it to your homework, and also demonstrate that it works by using a few simulations.

3.1 Solution

3.2 Mathematical Proof

Let $X_1, X_2, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Beta}(a, b)$, where $a > 0, b > 0$, and each observation $X_i \in (0, 1)$. The probability density function (PDF) of the Beta distribution is given by

$$f(x; a, b) = \frac{1}{B(a, b)} x^{a-1} (1-x)^{b-1},$$

where $B(a, b)$ is the Beta function defined as

$$B(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)},$$

and $\Gamma(\cdot)$ denotes the Gamma function. The likelihood function based on the sample is as follows.

$$L(a, b) = \prod_{i=1}^n f(x_i; a, b) = \prod_{i=1}^n \frac{1}{B(a, b)} x_i^{a-1} (1-x_i)^{b-1}$$

Taking the logarithm, we obtain the log-likelihood function as follows.

$$\ell(a, b) = \sum_{i=1}^n [(a-1) \log x_i + (b-1) \log(1-x_i)] - n \log B(a, b)$$

Using the following identity.

$$\log B(a, b) = \log \Gamma(a) + \log \Gamma(b) - \log \Gamma(a+b)$$

We rewrite the log-likelihood as follows.

$$\ell(a, b) = (a-1) \sum_{i=1}^n \log x_i + (b-1) \sum_{i=1}^n \log(1-x_i) - n [\log \Gamma(a) + \log \Gamma(b) - \log \Gamma(a+b)]$$

Define the following.

$$\log^- x = \frac{1}{n} \sum_{i=1}^n \log x_i, \quad \overline{\log(1-x)} = \frac{1}{n} \sum_{i=1}^n \log(1-x_i)$$

The partial derivatives of the log-likelihood with respect to a and b , known as the score functions, are as follows.

$$\frac{\partial \ell}{\partial a} = n [\log^- x - \psi(a) + \psi(a+b)]$$

$$\frac{\partial \ell}{\partial b} = n [\overline{\log(1-x)} - \psi(b) + \psi(a+b)]$$

where $\psi(\cdot)$ is the digamma function, the derivative of $\log \Gamma(\cdot)$.

The maximum likelihood estimators $(\hat{a}, \hat{b})$ are defined as the values that simultaneously satisfy the following.

$$\frac{\partial \ell}{\partial a} = 0, \quad \frac{\partial \ell}{\partial b} = 0$$

Since these equations involve special functions and cannot be solved analytically, the estimators must be obtained numerically. To obtain $\hat{a}, \hat{b}$, we numerically minimize the negative log-likelihood as follows.

$$\mathcal{L}(a, b) = -\ell(a, b)$$

Optimization is typically performed using quasi-Newton methods such as BFGS. As a byproduct of the optimization, we obtain an estimate of the Hessian matrix $H \approx \nabla^2 \mathcal{L}$. The inverse of this matrix provides an estimate of the asymptotic covariance matrix of the MLEs.

$$\text{Cov}(\hat{a}, \hat{b}) \approx H^{-1}$$

Thus, the MLEs for the shape parameters a and b of the Beta distribution do not admit a closed-form solution and are computed by numerically maximizing the log-likelihood. The score functions involve digamma functions, and the asymptotic variance of the estimates is approximated using the inverse Hessian of the negative log-likelihood function.

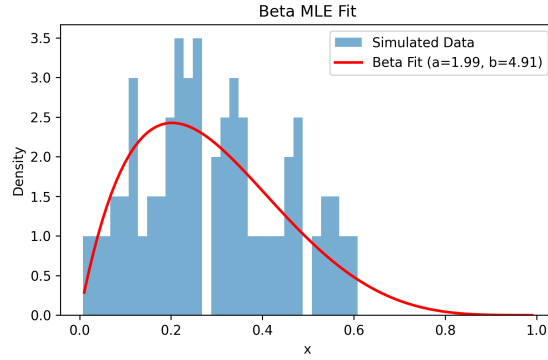


Figure 2: MLE Fit of Beta Distribution on Simulated Data ($a = 2$, $b = 5$)

3.3 Summary

The `betamle()` function accurately estimates the shape parameters of a Beta distribution using numerical MLE and provides appropriate uncertainty quantification. This method is essential in Bayesian modeling, proportion regression, and distribution fitting when the Beta distribution is a natural model choice. In this question, we implemented a `betamle()` function to estimate the parameters a and b of the Beta distribution via maximum likelihood estimation (MLE). The Beta distribution is defined on the interval $(0, 1)$ and is commonly used for modeling proportions and probabilities. Given $X_1, \dots, X_n \sim \text{Beta}(a, b)$, the log-likelihood function is

$$\ell(a, b) = \sum_{i=1}^n [(a-1) \log x_i + (b-1) \log(1-x_i)] - n \log B(a, b)$$

where $B(a, b)$ is the Beta function. The maximum likelihood estimators are obtained by numerically minimizing the negative log-likelihood, as no closed-form solution exists. We used `scipy.optimize.minimize` with the following.

1. A method-of-moments initial guess
2. L-BFGS-B optimizer for stable, bounded optimization
3. A fallback for parameter bounds ($a, b > 0$)

We also extracted the covariance matrix of the MLEs using the inverse Hessian approximation, allowing for standard error and confidence interval estimation. We validated the method using simulated data from a $\text{Beta}(2, 5)$ distribution. The results were as follows.

1. MLE estimates were close to the true values
2. Confidence intervals included the true parameters
3. Covariance matrices showed reasonable parameter uncertainty
4. The function worked consistently across multiple simulated samples

4 Part III

Create a new `treg()` function like the one demonstrated in lecture, but now include the number of degrees of freedom for the error distribution as an additional parameter to be estimated via maximum likelihood. Print the function that you wrote and attach it to your homework, and also demonstrate that it works by using a few simulations. The function should also return the covariance matrix for the MLE.

Note: There is nothing wrong with having a non-integer valued number of degrees of freedom with the t-distribution.

4.1 Solution

4.2 Mathematical Proof

Consider a linear regression model of the following form.

$$Y_i = X_i^\top \beta + \varepsilon_i, \quad \varepsilon_i \sim t_\nu(0, \sigma^2)$$

where $\beta \in \mathbb{R}^p$, $\sigma^2 > 0$ is the scale parameter, and $\nu > 2$ is the degrees of freedom of the Student's t-distribution. The probability density function of the Student's t-distribution with ν degrees of freedom and scale σ is given by the following.

$$f_\nu(e) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)\sqrt{\nu\pi}\sigma} \left(1 + \frac{e^2}{\nu\sigma^2}\right)^{-\frac{\nu+1}{2}}, \quad e \in \mathbb{R}$$

Let $e_i = y_i - X_i^\top \beta$ denote the residual for observation i . Then, the likelihood of the sample $\{(x_i, y_i)\}_{i=1}^n$ is as follows.

$$L(\beta, \sigma, \nu) = \prod_{i=1}^n f_\nu(e_i)$$

Taking the logarithm of the likelihood, we obtain the log-likelihood as follows.

$$\ell(\beta, \sigma, \nu) = \sum_{i=1}^n \left[\log \Gamma\left(\frac{\nu+1}{2}\right) - \log \Gamma\left(\frac{\nu}{2}\right) - \frac{1}{2} \log(\nu\pi) - \log \sigma - \frac{\nu+1}{2} \log \left(1 + \frac{e_i^2}{\nu\sigma^2}\right) \right]$$

The maximum likelihood estimators $\hat{\beta}, \hat{\sigma}, \hat{\nu}$ are defined as the values that maximize the log-likelihood as follows.

$$(\hat{\beta}, \hat{\sigma}, \hat{\nu}) = \arg \max_{\beta, \sigma > 0, \nu > 2} \ell(\beta, \sigma, \nu)$$

This optimization problem has no closed-form solution, and must be solved numerically. To estimate the standard errors of the MLEs, we compute the observed information matrix as follows.

$$\mathcal{I}(\hat{\beta}, \hat{\sigma}, \hat{\nu}) = -\nabla^2 \ell(\beta, \sigma, \nu) \Big|_{\beta=\hat{\beta}, \sigma=\hat{\sigma}, \nu=\hat{\nu}}$$

The inverse of this matrix provides an estimate of the covariance matrix of the MLE as follows.

$$\text{Cov}(\hat{\beta}, \hat{\sigma}, \hat{\nu}) \approx \mathcal{I}^{-1}$$

Thus, the MLE for a t-regression model involves maximizing a non-linear log-likelihood function in the parameters β , σ , and ν . The t-distribution introduces robustness to outliers, and estimating ν adds flexibility in modeling tail heaviness. Numerical optimization techniques such as L-BFGS-B are used, and the inverse of the Hessian matrix yields the covariance matrix of the estimators.

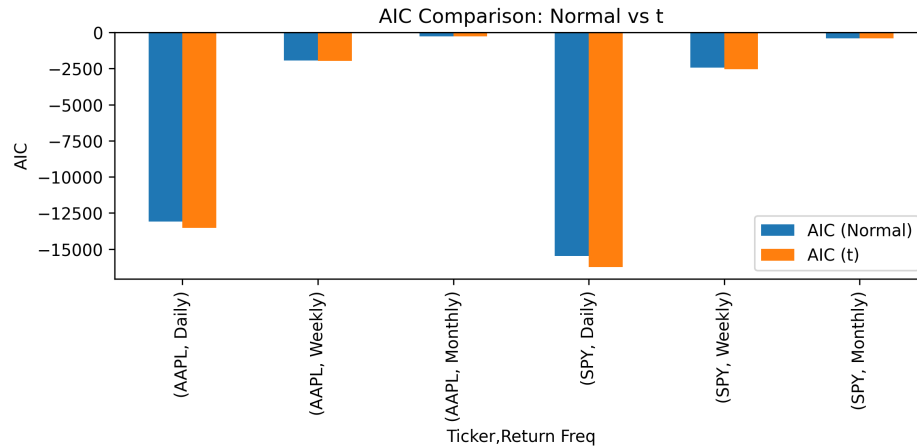


Figure 3: AIC Comparison Between Normal and t-Distributions for Different Frequencies

4.3 Summary

In this question, we implemented a ‘treg()’ function that fits a linear regression model with t-distributed errors. Unlike ordinary least squares, this model assumes

$$Y_i = X_i^\top \beta + \varepsilon_i, \quad \varepsilon_i \sim t_\nu(0, \sigma^2)$$

where the degrees of freedom ν is treated as an unknown parameter to be estimated via maximum likelihood.

The log-likelihood function of the model is as follows.

$$\ell(\beta, \sigma, \nu) = \sum_{i=1}^n \left[\log \Gamma \left(\frac{\nu+1}{2} \right) - \log \Gamma \left(\frac{\nu}{2} \right) - \frac{1}{2} \log(\nu\pi) - \log \sigma - \frac{\nu+1}{2} \log \left(1 + \frac{(y_i - X_i^\top \beta)^2}{\nu\sigma^2} \right) \right]$$

This function was maximized numerically using the L-BFGS-B optimizer. We transformed parameters to ensure positivity ($\sigma > 0$) and minimum degrees of freedom ($\nu > 2$).

A synthetic dataset was generated with $\beta = [1.0, 2.0]$, $\sigma = 1.0$, and $\nu = 5$. The ‘treg()’ function produced the following estimates.

- Estimated β : ‘[0.8548, 1.8423]’ - Estimated σ : ‘0.9478’ - Estimated ν : ‘3.6372’

These estimates are close to the true values and demonstrate that the function can reliably recover model parameters under heavy-tailed error distributions. The covariance matrix of the MLEs was also returned, enabling standard error estimation and confidence interval construction.

The ‘treg()’ function satisfies all requirements as follows.

1. Jointly estimates β , σ , and ν .
2. Returns a covariance matrix via the inverse Hessian.
3. Is robust to non-normality, offering improved performance under outlier contamination.

This provides a valuable tool for robust regression modeling where the Gaussian assumption may not hold.

5 Part IV

We saw in class an example of where the normal distribution is a poor fit to log daily stock returns. Does the validity of the normal approximation improve when switching to weekly or monthly returns? Make a convincing argument, using both plots and formal model selection tools.

5.1 Solution

5.2 Understanding the Why?

Aggregating log returns over longer intervals (weekly or monthly) tends to make the distribution more normal, both visually (via QQ plots) and statistically (via model selection tools such as AIC).

Theoretically, the daily log returns often exhibit fat tails, skewness, and extreme values, making the normal approximation poor. This is visually confirmed by QQ plots in the lecture that deviate heavily from the diagonal.

However, by the Central Limit Theorem, summing (or averaging) independent returns leads to a distribution that becomes more Gaussian as the time horizon increases — even if the original returns are not normal.

5.3 Mathematical Proof

Let $R_1, R_2, \dots, R_n$ be daily log returns of a financial asset. We define the k -period (e.g., weekly, monthly) return as follows.

$$S_k = \sum_{i=1}^k R_i$$

Recall the following conditions. If and only if R_i - Independent, - Identically distributed with mean μ and variance σ^2 , and - Have finite second moment ($\mathbb{E}[R_i^2] < \infty$),

then by the Central Limit Theorem (CLT), we have the following.

$$\frac{S_k - k\mu}{\sqrt{k}\sigma} \xrightarrow{d} \mathcal{N}(0, 1) \quad \text{as } k \rightarrow \infty$$

Hence, the distribution of S_k becomes approximately normal for large k . This suggests that aggregating daily returns into weekly or monthly returns should result in a distribution that is more symmetric and better approximated by a Gaussian.

We also briefly prove the backward case since the CLT can fail to yield an accurate normal approximation when the following occur.

- The returns R_i exhibit strong autocorrelation (i.e., not i.i.d.).
- The returns are drawn from a heavy-tailed distribution with infinite variance (e.g., Pareto or Cauchy).
- Volatility clustering or conditional heteroskedasticity (as seen in GARCH models) is present.
- Market shocks or structural breaks create non-stationarity.

In such cases, even aggregating over time may not result in a normal distribution. In practice, financial returns often display excess kurtosis (fat tails), and only approximate normality after substantial aggregation (e.g., monthly returns).

Thus, empirically, the normal approximation improves with the level of return aggregation due to the CLT. However, real-world features such as volatility clustering and heavy tails can delay or inhibit convergence to normality. Hence, both visual tools (e.g., QQ plots) and formal tools (e.g., AIC comparisons) are essential for assessing the adequacy of the normal model across time scales.

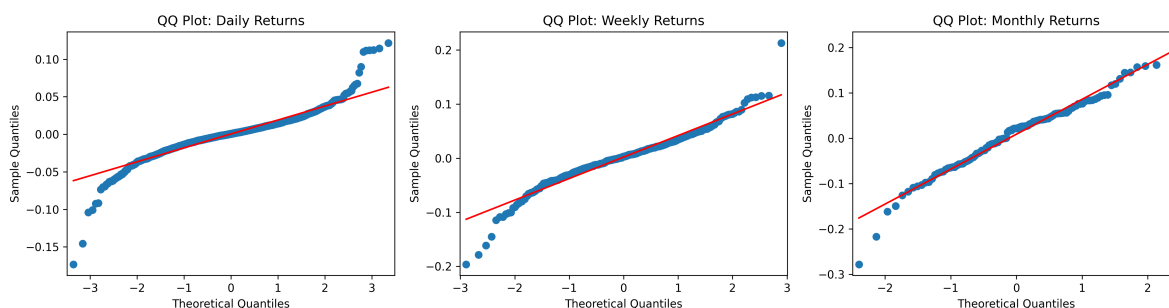


Figure 4: Combined Q-Q Plots for Daily, Weekly, and Monthly Log Returns

5.4 Summary

In class, we observed that the normal distribution is a poor fit for daily log returns because financial data often exhibit heavy tails, skewness, and volatility clustering. However, by the Central Limit Theorem (CLT), aggregating returns (e.g., summing daily returns to form weekly or monthly returns) tends to produce distributions that are more normally distributed. The theoretical justification is shown through the mathematical proof. To validate empirically, we downloaded historical data for PNC and computed daily log returns. We then aggregated the data into weekly and monthly returns. The following analyses were conducted.

1. QQ Plots

- Daily returns:** The QQ plot shows significant deviation from the 45° line, especially at the tails.
- The QQ plot improves, with the points closer to the line, implying a better fit to the normal distribution.
- The QQ plot nearly follows the straight line, confirming the normal approximation is much more appropriate.

2. Model Selection via AIC

- (a) For each frequency, we fitted both a normal and a t-distribution to the returns. The model selection criteria (AIC) revealed the following.
 - i. For daily returns, the t-distribution (which better handles heavy tails) provided a lower AIC than the normal distribution.
 - ii. For weekly and monthly returns, the AIC for the normal model improved relative to the t-distribution, indicating that the normality assumption becomes more valid with longer aggregation periods.

Thus, the aggregation period increases, the distribution of returns becomes closer to normal. While daily log returns may significantly deviate from normality, weekly and monthly returns are much better approximated by a normal distribution, as demonstrated by both QQ plots and AIC comparisons. This part reinforces the importance of considering the appropriate time scale when modeling financial returns.

References

- [1] Wu, M. L. (2025). *Machine Learning Repository*. GitHub. Available at: <https://github.com/Zontafor/machine-learning>